Hyperspectral Target Detection Using a Bilinear Sparse Binary Hypothesis Model

Yanzi Shi, *Graduate Student Member, IEEE*, Jiaojiao Li, *Member, IEEE*, Yunsong Li, *Member, IEEE*, and Paolo Gamba, *Fellow, IEEE*

***Abstract*— The binary hypothesis testing (BHT) is one of the most important models in hyperspectral target detection (HTD). However, this model is generally based on a linear mixture model (LMM) and might be inaccurate to reflect target and background characterizations in some scenes. This article presents a bilinear sparse target detector (BSTD) by applying the bilinear sparse mixture model (BSMM) to a popular BHT-based detection algorithm termed adaptive matched subspace detector (AMSD), which takes bilinear target–background interaction and sparse abundance into account. Moreover, as AMSD relies heavily on background subspace, we design a robust background subspace construction method. Specifically, we first classify each pixel into noise, border, or other particular instances according to its density, which is measured by jointly spatial–spectral distance. With the coarse classification map, a class-guided automatic background generation (CABG) process is introduced to reliably generate pure background samples. Detection statistics and component analysis on five real-world hyperspectral images verify the effectiveness of our BSTD method.**

***Index Terms*— Bilinear sparse target detection, class-guided automatic background generation (CABG), hyperspectral imagery, spatial–spectral density-based classification.**

I. Introduction

SIGNIFICANT progress has been made in the use of hyperspectral remote sensing techniques to recognize various substances of interest in the past few decades. Such technique is termed hyperspectral target detection (HTD), which estimates the probability that the interested target prior is present in each test pixel and has held the researchers' attention in many engineering applications, such as Earth observation [1], deep space exploration [2], and health hazards recognition [3].

There are many classical and superior algorithms inferred from binary hypothesis testing (BHT) to achieve precise detection in hyperspectral imagery, including the spectral matched filter (SMF) [4], the adaptive coherence/cosine estimator (ACE), also known as the adaptive subspace detector (ASD) [5], the adaptive matched subspace detector (AMSD) [6], the recent sparsity-based target detector (STD) [7], the combined sparse and collaborative representation-based method (CSCR) [8], and their improved versions, such as kernel SMF (KSMF) [4] and the sparse representation-based binary hypothesis target detector (SRBBHTD) [9]. For clarity, Table I lists the formulation of the aforementioned HTD algorithms. These methods are all based on the linear mixture model (LMM), which suits scenes where no interactions among different instances occur. Note that, although the sparse representation-based methods have a similar model to the subspace detection methods, they make no assumptions on noise/background distribution. On the other hand, for some complex scenarios, scattering among multiple instances is quite common, and LMM cannot characterize this situation well. This view has been presented and demonstrated in many papers about hyperspectral unmixing [10], [11].

To address this issue, several nonlinear mixture models (NLMMs) have been proposed, such as radiative transfer-based approaches, kernel methods, bilinear models, p-order polynomial models, and neural networks. Directly deriving spectral signatures and their abundance from the radiative transfer equation (RTE) is extremely complicated, and some simplified models, such as Hapke [14] and Shkuratov *et al.* [15], have been proposed to approximate RTE. In order to avoid complex RTE models, Broadwater and Banerjee [16] exploited the kernel trick, including physics-based kernels and other popular kernel functions (e.g., linear, polynomial, and radial basis) to unmix HSIs. Besides that bilinear mixture models (BMMs) extended from LMM also achieve outstanding unmixing performance, typical methods include Nascimento and Bioucas-Dias [17], Fan *et al.* [18], generalized BMM (GBM) [19], and polynomial postnonlinear mixing model (PPNMM) [20]. The p-order polynomial NLMM further models higher order interactions and also provides a framework for efficient parallel computation [21]. In recent years, data-driven neural networks have

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Yanzi Shi, Jiaojiao Li, and Yunsong Li are with the State Key Laboratory of Integrated Service Network, Xidian University, Xi'an 710071, China (e-mail: yzshi_xidian@163.com; jjli@xidian.edu.cn; ysli@mail.xidian.edu.cn).

Paolo Gamba is with the Department of Electrical, Computer and Biomedical Engineering, University of Pavia, 27100 Pavia, Italy (e-mail: paolo.gamba@unipv.it).

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TABLE I
FORMULATION OF SEVERAL POPULAR BINARY HYPOTHESIS TESTING-BASED HTD ALGORITHMS

Method	Hypothesis 0 (H_0)	Hypothesis 1 (H_1)	Constraint	Detector $D(\mathbf{x})$
SMF	$\mathbf{x}=\mathbf{n}$	$\mathbf{x}=\alpha_t\mathbf{x}_t+\mathbf{n}$	-	$\frac{\mathbf{x}_t^T\hat{\mathbf{C}}^{-1}\mathbf{x}}{\mathbf{x}_t^T\hat{\mathbf{C}}^{-1}\mathbf{x}_t}$
KSMF	$\Phi(\mathbf{x})=\mathbf{n}_\Phi$	$\Phi(\mathbf{x})=\alpha_\Phi^t\mathbf{X}_\Phi^t+\mathbf{n}_\Phi$	-	$\frac{\Phi(\mathbf{x}_\Phi^t)^T\hat{\mathbf{C}}_\Phi^{-1}\Phi(\mathbf{x})}{\Phi(\mathbf{x}_\Phi^t)^T\hat{\mathbf{C}}_\Phi^{-1}\Phi(\mathbf{x}_\Phi^t)}$
ACE/ASD	$\mathbf{x}=\mathbf{n}$	$\mathbf{x}=\mathbf{X}_t\boldsymbol{\alpha}_t+\sigma\mathbf{n}$	-	$\frac{\mathbf{x}^T\hat{\mathbf{C}}^{-1}\mathbf{X}_t\left(\mathbf{X}_t^T\hat{\mathbf{C}}^{-1}\mathbf{X}_t\right)^{-1}\mathbf{X}_t^T\hat{\mathbf{C}}^{-1}\mathbf{x}}{\mathbf{x}^T\hat{\mathbf{C}}^{-1}\mathbf{x}}$
KASD [5]	$\Phi(\mathbf{x})=\mathbf{n}_\Phi$	$\Phi(\mathbf{x})=\mathbf{X}_\Phi^t\boldsymbol{\alpha}_\Phi^t+\sigma_\Phi\mathbf{n}_\Phi$	-	$\frac{\Phi(\mathbf{x})^T\hat{\mathbf{C}}_\Phi^{-1}\mathbf{X}_\Phi^t\left(\mathbf{X}_\Phi^{t\,T}\hat{\mathbf{C}}_\Phi^{-1}\mathbf{X}_\Phi^t\right)^{-1}\mathbf{X}_\Phi^{t\,T}\hat{\mathbf{C}}_\Phi^{-1}\Phi(\mathbf{x})}{\Phi(\mathbf{x})^T\hat{\mathbf{C}}_\Phi^{-1}\Phi(\mathbf{x})}$
MSD [4]	$\mathbf{x}=\mathbf{X}_b\boldsymbol{\alpha}_{b,0}+\mathbf{n}$	$\mathbf{x}=\mathbf{X}_t\boldsymbol{\alpha}_t+\mathbf{X}_b\boldsymbol{\alpha}_{b,1}+\mathbf{n}$	-	$\frac{\mathbf{x}^T\left(\mathbf{I}-\mathbf{X}_b\left(\mathbf{X}_b^T\mathbf{X}_b\right)^{-1}\mathbf{X}_b^T\right)\mathbf{x}}{\mathbf{x}^T\left(\mathbf{I}-\mathbf{X}_{tb}\left(\mathbf{X}_{tb}^T\mathbf{X}_{tb}\right)^{-1}\mathbf{X}_{tb}^T\right)\mathbf{x}}$
KMSD [12]	$\Phi(\mathbf{x})=\mathbf{X}_\Phi^b\boldsymbol{\alpha}_\Phi^{b,0}+\mathbf{n}_\Phi$	$\Phi(\mathbf{x})=\mathbf{X}_\Phi^b\boldsymbol{\alpha}_\Phi^{b,1}+\mathbf{X}_\Phi^t\boldsymbol{\alpha}_\Phi^t+\mathbf{n}_\Phi$	-	$\frac{\Phi(\mathbf{x})^T\left(\mathbf{I}_\Phi-\mathbf{X}_\Phi^b\left(\mathbf{X}_\Phi^{b\,T}\mathbf{X}_\Phi^b\right)^{-1}\mathbf{X}_\Phi^{b\,T}\right)\Phi(\mathbf{x})}{\Phi(\mathbf{x})^T\left(\mathbf{I}_\Phi-\mathbf{X}_\Phi^{tb}\left(\mathbf{X}_\Phi^{tb\,T}\mathbf{X}_\Phi^{tb}\right)^{-1}\mathbf{X}_\Phi^{tb\,T}\right)\Phi(\mathbf{x})}$
AMSD	$\mathbf{x}=\mathbf{X}_b\boldsymbol{\alpha}_{b,0}+\mathbf{n}_b$	$\mathbf{x}=\mathbf{X}_t\boldsymbol{\alpha}_t+\mathbf{X}_b\boldsymbol{\alpha}_{b,1}+\mathbf{n}_{tb}$	-	$\frac{\mathbf{x}^T\left(\mathbf{I}-\mathbf{X}_b\left(\mathbf{X}_b^T\mathbf{X}_b\right)^{-1}\mathbf{X}_b^T\right)\mathbf{x}}{\mathbf{x}^T\left(\mathbf{I}-\mathbf{X}_{tb}\left(\mathbf{X}_{tb}^T\mathbf{X}_{tb}\right)^{-1}\mathbf{X}_{tb}^T\right)\mathbf{x}}$
STD	$\mathbf{x}=\mathbf{D}_b\boldsymbol{\alpha}_b+\mathbf{e}_0$	$\mathbf{x}=\mathbf{D}_t\boldsymbol{\alpha}_t+\mathbf{e}_1$	$\left[\begin{smallmatrix}\boldsymbol{\alpha}_b\\ \boldsymbol{\alpha}_t\end{smallmatrix}\right]$ is sparse.	$\|\mathbf{x}-\mathbf{D}_b\hat{\boldsymbol{\alpha}}_b\|_2-\|\mathbf{x}-\mathbf{D}_t\hat{\boldsymbol{\alpha}}_t\|_2$
SRBBHTD	$\mathbf{x}=\mathbf{D}_b\boldsymbol{\alpha}_{b,0}+\mathbf{e}_0$	$\mathbf{x}=\mathbf{D}_t\boldsymbol{\alpha}_t+\mathbf{D}_b\boldsymbol{\alpha}_{b,1}+\mathbf{e}_1$	$\boldsymbol{\alpha}_{b,0}$ and $\left[\begin{smallmatrix}\boldsymbol{\alpha}_b^1\\ \boldsymbol{\alpha}_t\end{smallmatrix}\right]$ are sparse.	$\|\mathbf{x}-\mathbf{D}_b\hat{\boldsymbol{\alpha}}_{b,0}\|-\|\mathbf{x}-\mathbf{D}_t\hat{\boldsymbol{\alpha}}_t-\mathbf{D}_b\hat{\boldsymbol{\alpha}}_{b,1}\|$
KSRBBHTD [13]	$\Phi(\mathbf{x})\approx\mathbf{D}_\Phi^b\boldsymbol{\alpha}_\Phi^{b,0}$	$\Phi(\mathbf{x})\approx\mathbf{D}_\Phi^b\boldsymbol{\alpha}_\Phi^{b,1}+\mathbf{D}_\Phi^t\boldsymbol{\alpha}_\Phi^t$	$\boldsymbol{\alpha}_\Phi^{b,0}$ and $\left[\begin{smallmatrix}\boldsymbol{\alpha}_\Phi^{b,1}\\ \boldsymbol{\alpha}_\Phi^t\end{smallmatrix}\right]$ are sparse.	$\left\|\Phi(\mathbf{x})-\mathbf{D}_\Phi^b\hat{\boldsymbol{\alpha}}_\Phi^{b,0}\right\|-\left\|\Phi(\mathbf{x})-\mathbf{D}_\Phi^t\hat{\boldsymbol{\alpha}}_\Phi^t-\mathbf{D}_\Phi^b\hat{\boldsymbol{\alpha}}_\Phi^{b,1}\right\|$
CSCR	$\mathbf{x}=\mathbf{D}_b\boldsymbol{\alpha}_b$	$\mathbf{x}=\mathbf{D}_t\boldsymbol{\alpha}_t$	$\boldsymbol{\alpha}_t$ is sparse, $\boldsymbol{\alpha}_b$ is collaborative.	$\|\mathbf{x}-\mathbf{D}_b\hat{\boldsymbol{\alpha}}_b\|_2^2-\|\mathbf{x}-\mathbf{D}_t\hat{\boldsymbol{\alpha}}_t\|_2^2$

Note: $\mathbf{X}_t$, $\mathbf{X}_b$, $\mathbf{D}_t$, and $\mathbf{D}_b$ represent target subspace, background subspace, target dictionary, and background dictionary, respectively. $\boldsymbol{\alpha}_t$ and $\boldsymbol{\alpha}_b$ are coefficients corresponding to target and background. $\mathbf{n}$ and $\mathbf{e}$ denote noise and approximation residual, respectively, and $\mathbf{n}$ is assumed to comply with a particular distribution. $\#_\phi$ is a representation of $\#$ in the high-dimensional feature space.

been successively applied to nonlinear unmixing due to their strong ability of nonlinear feature extraction [22], [23]. Existing studies in [18], [19], [24], [25] on hyperspectral unmixing have verified the superiority of NLMM over LMM for complex scenes.

As for HTD, Wang and Xue [26] proposed a hyperspectral target detector termed matched subspace detector with interaction effects (MSDinter), which integrates classical target detectors with the nonlinear interactions between target and background spectra. To the best of our knowledge, this is the first work that applies NLMM to HTD, and it also proves the advantages of taking nonlinear interactions between target and background into account. Later, Yang *et al.* [27] improved MSDinter by augmenting scarce target samples to construct more reliable target and background subspaces. In addition to nonlinear HTD by introducing NLMM, several deep learning-based HTD algorithms have been presented to capture nonlinear properties of HSI, in which CNN-based detection (CNND) [28], CNN target detector (CNNTD) [29], HTD-Net [30], and two-stream convolutional network-based target detector (TSCNTD) [31] are performed in a supervised way, and the distance constrained stacked sparse autoencoder-based detector (DCSSAED) [32], the 3-D macro–micro-residual autoencoder detector (3D-MMRAED) [33], and the deep spatial–spectral network (DSSN) [34] used unsupervised networks. Although these deep learning-based HTD methods generally obtain better detection performance than typical statistical methods, they are severely constrained by expensive computation complexity and sensor-specific transferability.

In this article, we propose a new bilinear sparse target detector (BSTD) for hyperspectral imagery, which upgrades LMM in AMSD to the bilinear sparse mixture model (BSMM), i.e., introducing nonlinear target–background interactions modeled with simple BMM and sparse target abundance constrained by ℓ_1-norm into AMSD. In addition to accurate mixture models, it shows that background subspace has a significant impact on the detection performance of AMSD. To build a valid background subspace, we present a class-guided automatic background generation (CABG) method, which uses a spatial–spectral density-based classification map to guide the automatic target generation process (ATGP) [35]. Compared with the classical ATGP, it can exclude noisy and boundary pixels from background candidates and, thus, help to extract more pure background samples.

The remainder of this article is organized as follows. Section II shortly summarizes BMM and AMSD. Section III describes the bilinear sparse binary hypothesis detection model and robust background subspace construction in detail. The detection statistics and component analysis on five real-world hyperspectral datasets are provided in Section IV. Section V draws the conclusions of this article.

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TABLE II
SEVERAL POPULAR BMM-BASED MIXTURE MODELS UNDER DIFFERENT WAYS OF PARAMETER SETUP

Model	$\beta^{i,j}$	n_1	n_2
Nascimento	$\beta^{i,j}$	n_e-1	$i+1$
Fan	$\boldsymbol{\alpha}^i\boldsymbol{\alpha}^j$	n_e-1	$i+1$
GBM	$\gamma^{i,j}\boldsymbol{\alpha}^i\boldsymbol{\alpha}^j$	n_e-1	$i+1$
PPNMM	$\gamma\boldsymbol{\alpha}^i\boldsymbol{\alpha}^j$	n_e	1

II. Related Works

A. Bilinear Mixture Model

In comparison with LMM, the bilinear mixture model (BMM) additionally introduces a bilinear interaction term to model second-order scattering, formulated as

$$\mathbf{x}=\underbrace{\mathbf{X}_e\boldsymbol{\alpha}}_{\text{linear}}+\underbrace{\sum_{i=1}^{n_1}\sum_{j=n_2}^{n_e}\beta^{i,j}\mathbf{x}_e^i\odot\mathbf{x}_e^j}_{\text{bilinear}}+\underbrace{\mathbf{n}}_{\text{noise}} \tag{1}$$

where $\mathbf{X}_e=[\mathbf{x}_e^1\ \mathbf{x}_e^2\ \ldots\ \mathbf{x}_e^{n_e}]\in\mathbb{R}^{n_l\times n_e}$ denotes n_e endmembers with n_l spectral bands, and $\boldsymbol{\alpha}\in\mathbb{R}^{n_e\times 1}$ and $\beta^{i,j}$ are linear and bilinear coefficients, respectively. "$\odot$" denotes the elementwise (Hadamard) product. Different $\beta^{i,j}$'s, n_1's, and n_2's generate different BMMs. Table II gives several popular BMM-based mixture models.

The aforementioned models are designed for hyperspectral unmixing in particular and should satisfy sum-to-one and nonnegative constraints to preserve physical significance. However, HTD only pays attention to the existence of the interested target rather than the exact absolute abundance. Therefore, these two constraints are usually ignored to simplify the detection process.

B. Adaptive Matched Subspace Detector (AMSD)

AMSD is derived from BHT. Given an HSI $\mathbf{X}\in\mathbb{R}^{n_h\times n_w\times n_l}$, the BHT model for each test sample $\mathbf{x}^k\in\mathbb{R}^{n_l\times 1}$ ($k=1,2,\ldots,n_h\times n_w$) can be expressed as

$$\begin{aligned}H_0\text{: }&\mathbf{x}^k=\mathbf{X}_b\boldsymbol{\alpha}_{b,0}^k+\mathbf{n}_b,\quad\text{target absent}\\H_1\text{: }&\mathbf{x}^k=\mathbf{X}_t\boldsymbol{\alpha}_t^k+\mathbf{X}_b\boldsymbol{\alpha}_{b,1}^k+\mathbf{n}_{tb},\quad\text{target present}\end{aligned} \tag{2}$$

in which $\mathbf{X}_b\in\mathbb{R}^{n_l\times n_b}$ and $\mathbf{X}_t\in\mathbb{R}^{n_l\times n_t}$ represent orthogonal background subspace and target subspace, respectively, i.e., $\mathbf{X}_b^T\mathbf{X}_b=\mathbf{I}_{n_b}$ and $\mathbf{X}_t^T\mathbf{X}_t=\mathbf{I}_{n_t}$. The vectors $\boldsymbol{\alpha}_{b,0}^k$ and $\begin{bmatrix}\boldsymbol{\alpha}_t^k\\ \boldsymbol{\alpha}_{b,1}^k\end{bmatrix}$ are linear coefficients under H_0 and H_1 hypotheses, respectively. $\mathbf{n}_b\sim\mathcal{N}(0,\sigma_b^2\mathbf{I}_{n_l})$ and $\mathbf{n}_{tb}\sim\mathcal{N}(0,\sigma_{tb}^2\mathbf{I}_{n_l})$ denote the Gaussian noise with different covariances. If $\mathbf{x}^k$ is a background sample, it could be represented as a linear combination of pure background bases with small error. Otherwise, if $\mathbf{x}^k$ fully or partially belongs to target, it cannot be well reconstructed unless the target bases are additionally introduced.

The generalized likelihood ratio test (GLRT) is used to perform detection statistics. Let $p(\cdot)$ denote the probability density function, $\boldsymbol{\alpha}_{tb}^k=\begin{bmatrix}\boldsymbol{\alpha}_t^k\\ \boldsymbol{\alpha}_{b,1}^k\end{bmatrix}$, and $\mathbf{X}_{tb}=[\mathbf{X}_t\ \mathbf{X}_b]$; then, AMSD can be formulated as

$$\begin{aligned}D_{\text{AMSD}}(\mathbf{x}^k)&=\frac{p(\hat{\boldsymbol{\alpha}}_{tb}^k,\hat{\sigma}_{tb}^2;\mathbf{x}^k)}{p(\hat{\boldsymbol{\alpha}}_{b,0}^k,\hat{\sigma}_b^2;\mathbf{x}^k)}\\&=\left(\frac{\hat{\sigma}_b^2}{\hat{\sigma}_{tb}^2}\right)^{\frac{n_l}{2}}\exp\left\{-\frac{1}{2\hat{\sigma}_{tb}^2}\|\hat{\mathbf{n}}_{tb}\|_2^2+\frac{1}{2\hat{\sigma}_b^2}\|\hat{\mathbf{n}}_b\|_2^2\right\}\end{aligned} \tag{3}$$

where $\hat{\sigma}_b^2=(1/n_l)\|\mathbf{n}_b\|_2^2$ and $\hat{\sigma}_{tb}^2=(1/n_l)\|\mathbf{n}_{tb}\|_2^2$. For simplicity, let us omit the exponent of $(\hat{\sigma}_b^2/\hat{\sigma}_{tb}^2)^{(n_l/2)}$, and (3) becomes

$$D_{\text{AMSD}}(\mathbf{x}^k)=\frac{\hat{\sigma}_b^2}{\hat{\sigma}_{tb}^2}=\frac{\|\hat{\mathbf{n}}_b\|_2^2}{\|\hat{\mathbf{n}}_{tb}\|_2^2}=\frac{\left\|\mathbf{x}^k-\mathbf{X}_b\hat{\boldsymbol{\alpha}}_{b,0}^k\right\|_2^2}{\left\|\mathbf{x}^k-\mathbf{X}_{tb}\hat{\boldsymbol{\alpha}}_{tb}^k\right\|_2^2}. \tag{4}$$

The maximum likelihood estimations (MLEs) of $\boldsymbol{\alpha}_{b,0}^k$ and $\boldsymbol{\alpha}_{tb}^k$ could be found by solving the following two optimization problems:

$$\hat{\boldsymbol{\alpha}}_{b,0}^k=\arg\max_{\boldsymbol{\alpha}_{b,0}^k}\left\{-\frac{\left\|\mathbf{x}_k-\mathbf{X}_b\boldsymbol{\alpha}_{b,0}^k\right\|_2^2}{2\sigma_b^2}\right\} \tag{5a}$$

$$\hat{\boldsymbol{\alpha}}_{tb}^k=\arg\max_{\boldsymbol{\alpha}_{tb}^k}\left\{-\frac{\left\|\mathbf{x}_k-\mathbf{X}_{tb}\boldsymbol{\alpha}_{tb}^k\right\|_2^2}{2\sigma_{tb}^2}\right\}. \tag{5b}$$

It is clear that the objective functions in (5a) and (5b) are quadratic with respect to parameters $\boldsymbol{\alpha}_{b,0}^k$ and $\boldsymbol{\alpha}_{tb}^k$. Therefore, by differentiating to $\boldsymbol{\alpha}_{b,0}^k$ and $\boldsymbol{\alpha}_{tb}^k$, we have

$$\hat{\boldsymbol{\alpha}}_{b,0}^k=\left(\mathbf{X}_b^T\mathbf{X}_b\right)^{-1}\mathbf{X}_b^T\mathbf{x}^k \tag{6a}$$

$$\hat{\boldsymbol{\alpha}}_{tb}^k=\left(\mathbf{X}_{tb}^T\mathbf{X}_{tb}\right)^{-1}\mathbf{X}_{tb}^T\mathbf{x}^k. \tag{6b}$$

Substituting (6a) and (6b) back into (4), AMSD becomes

$$\begin{aligned}&D_{\text{AMSD}}(\mathbf{x}^k)\\&=\frac{\left(\mathbf{x}^k-\mathbf{X}_b\left(\mathbf{X}_b^T\mathbf{X}_b\right)^{-1}\mathbf{X}_b^T\mathbf{x}^k\right)^T\left(\mathbf{x}^k-\mathbf{X}_b\left(\mathbf{X}_b^T\mathbf{X}_b\right)^{-1}\mathbf{X}_b^T\mathbf{x}^k\right)}{\left(\mathbf{x}^k-\mathbf{X}_{tb}\left(\mathbf{X}_{tb}^T\mathbf{X}_{tb}\right)^{-1}\mathbf{X}_{tb}^T\mathbf{x}^k\right)^T\left(\mathbf{x}^k-\mathbf{X}_{tb}\left(\mathbf{X}_{tb}^T\mathbf{X}_{tb}\right)^{-1}\mathbf{X}_{tb}^T\mathbf{x}^k\right)}\\&=\frac{{\mathbf{x}^k}^T\left(\mathbf{I}_{n_l}-\mathbf{X}_b\left(\mathbf{X}_b^T\mathbf{X}_b\right)^{-1}\mathbf{X}_b^T\right)\mathbf{x}^k}{{\mathbf{x}^k}^T\left(\mathbf{I}_{n_l}-\mathbf{X}_{tb}\left(\mathbf{X}_{tb}^T\mathbf{X}_{tb}\right)^{-1}\mathbf{X}_{tb}^T\right)\mathbf{x}^k}\\&=\frac{{\mathbf{x}^k}^T\left(\mathbf{I}_{n_l}-\mathbf{X}_b\mathbf{X}_b^T\right)\mathbf{x}^k}{{\mathbf{x}^k}^T\left(\mathbf{I}_{n_l}-\mathbf{X}_{tb}\left(\mathbf{X}_{tb}^T\mathbf{X}_{tb}\right)^{-1}\mathbf{X}_{tb}^T\right)\mathbf{x}^k}.\end{aligned} \tag{7}$$

If $D_{\text{AMSD}}(\mathbf{x}^k)>\eta$ (η is a threshold in correspondence with a specific false alarm rate), $\mathbf{x}^k$ is detected as target; otherwise, $\mathbf{x}^k$ is a background sample.

III. Proposed Method

In order to obtain better target detection performance in complex scenes, nonlinear interactions between multiple instances should be considered. This article characterizes these nonlinearities with BMM and also takes abundance sparsity into consideration, generating a BHT-based BSTD. In addition to the improved detection model, a dependable background subspace is also of great significance for HTD. To achieve this, we construct the background subspace on the premise

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of eliminating noisy and boundary-mixed pixels according to a density-based classification map. Details are given in Sections III-A and III-B.

A. Bilinear Sparse Target Detection

1) Problem Formulation: Directly introducing BMM in formula (1) (Nascimento is used for simplicity) into BHT in formula (2), the bilinear binary hypothesis model is defined as

$$\begin{aligned} H_0\text{: } \mathbf{x}^k &= \mathbf{X}_b\boldsymbol{\alpha}_{b,0}^k + \sum_{i=1}^{n_b-1}\sum_{j=i+1}^{n_b}\beta_b^{k,i,j}\mathbf{x}_b^i \odot \mathbf{x}_b^j + \mathbf{n}_b, \\ &\qquad\qquad \text{target absent} \\ H_1\text{: } \mathbf{x}^k &= \mathbf{X}_{tb}\boldsymbol{\alpha}_{tb}^k + \sum_{i=1}^{n_t+n_b-1}\sum_{j=i+1}^{n_t+n_b}\beta_{tb}^{k,i,j}\mathbf{x}_{tb}^i \odot \mathbf{x}_{tb}^j + \mathbf{n}_{tb}, \\ &\qquad\qquad \text{target present.} \end{aligned} \tag{8}$$

The AMSD detector in (4) can be regarded as ratio between the recovery residuals obtained by H_0 and H_1; thus, larger $\|\mathbf{x}^k - \mathbf{X}_b\hat{\boldsymbol{\alpha}}_{b,0}^k\|_2^2$ and smaller $\|\mathbf{x}^k - \mathbf{X}_{tb}\hat{\boldsymbol{\alpha}}_{tb}^k\|_2^2$ could produce a better detector. In this case, we model hypotheses H_0 and H_1 with slightly inaccurate LMM and more accurate BMM, respectively, and recast them as follows:

$$\begin{aligned} H_0\text{: } \mathbf{x}^k &= \mathbf{X}_b\boldsymbol{\alpha}_{b,0}^k + \mathbf{n}_b, \quad \text{target absent} \\ H_1\text{: } \mathbf{x}^k &= \mathbf{X}_{tb}\boldsymbol{\alpha}_{tb}^k + \sum_{i=1}^{n_t+n_b-1}\sum_{j=i+1}^{n_t+n_b}\beta_{tb}^{k,i,j}\mathbf{x}_{tb}^i \odot \mathbf{x}_{tb}^j + \mathbf{n}_{tb}, \\ &\qquad\qquad \text{target present} \end{aligned} \tag{9}$$

in which the amount of abundance parameters to be estimated under H_0 is reduced by $((n_b(n_b-1))/2)$, and $\sum_{i=1}^{n_t+n_b-1}\sum_{j=i+1}^{n_t+n_b}\mathbf{x}_{tb}^i \odot \mathbf{x}_{tb}^j$ can be decomposed into three parts as follows:

$$\begin{aligned} &\sum_{i=1}^{n_t+n_b-1}\sum_{j=i+1}^{n_t+n_b}\mathbf{x}_{tb}^i \odot \mathbf{x}_{tb}^j \\ &= \sum_{i=1}^{n_t-1}\sum_{j=i+1}^{n_t+n_b}\mathbf{x}_{tb}^i \odot \mathbf{x}_{tb}^j + \mathbf{x}_{tb}^{n_t} \odot \sum_{j=n_t+1}^{n_t+n_b}\mathbf{x}_{tb}^j \\ &\quad + \sum_{i=n_t+1}^{n_t+n_b-1}\sum_{j=i+1}^{n_t+n_b}\mathbf{x}_{tb}^i \odot \mathbf{x}_{tb}^j \\ &= \sum_{i=1}^{n_t-1}\sum_{j=i+1}^{n_t}\mathbf{x}_{tb}^i \odot \mathbf{x}_{tb}^j + \sum_{i=1}^{n_t-1}\sum_{j=n_t+1}^{n_t+n_b}\mathbf{x}_{tb}^i \odot \mathbf{x}_{tb}^j \\ &\quad + \mathbf{x}_{tb}^{n_t} \odot \sum_{j=n_t+1}^{n_t+n_b}\mathbf{x}_{tb}^j + \sum_{i=n_t+1}^{n_t+n_b-1}\sum_{j=i+1}^{n_t+n_b}\mathbf{x}_{tb}^i \odot \mathbf{x}_{tb}^j \\ &= \sum_{i=1}^{n_t-1}\sum_{j=i+1}^{n_t}\mathbf{x}_t^i \odot \mathbf{x}_t^j + \sum_{i=1}^{n_t-1}\sum_{j=1}^{n_b}\mathbf{x}_t^i \odot \mathbf{x}_b^j \\ &\quad + \mathbf{x}_t^{n_t} \odot \sum_{j=1}^{n_b}\mathbf{x}_b^j + \sum_{i=1}^{n_b-1}\sum_{j=i+1}^{n_b}\mathbf{x}_b^i \odot \mathbf{x}_b^j \\ &= \sum_{i=1}^{n_t-1}\sum_{j=i+1}^{n_t}\mathbf{x}_t^i \odot \mathbf{x}_t^j + \sum_{i=1}^{n_t}\sum_{j=1}^{n_b}\mathbf{x}_t^i \odot \mathbf{x}_b^j + \sum_{i=1}^{n_b-1}\sum_{j=i+1}^{n_b}\mathbf{x}_b^i \odot \mathbf{x}_b^j. \end{aligned} \tag{10}$$

For an imaged pixel, if it is composed of only target material, no bilinear interactions between backgrounds, i.e., the $\mathbf{x}_b^i \odot \mathbf{x}_b^j$ term, exist in this case; if it includes both target and background materials, discuss in two situations: if the target abundance is low abundance, meaning that the energy of target is quite small, the imaged pixel generally falls near background and is considered as background for simplicity; if the background abundance is low, the bilinear interactions between backgrounds contribute much less to the composite spectrum compared with the linear background mixture, and its removal has little impacts on the imaged pixel. Therefore, the last term $\sum_{i=1}^{n_b-1}\sum_{j=i+1}^{n_b}\mathbf{x}_b^i \odot \mathbf{x}_b^j$ in (10) could be neglected. In addition, fewer parameters are required to be calculated, especially in a scene with complex background. As only one target spectrum is assumed to be provided in most cases, hence, we let $n_t = 1$. Under these conditions, the bilinear binary hypothesis model translates into the following expressions:

$$\begin{aligned} H_0\text{: } \mathbf{x}^k &= \mathbf{X}_b\boldsymbol{\alpha}_{b,0}^k + \mathbf{n}_b, \quad \text{target absent} \\ H_1\text{: } \mathbf{x}^k &= \mathbf{X}_{tb}\boldsymbol{\alpha}_{tb}^k + \mathbf{x}_t \odot \sum_{j=1}^{n_b}\beta_{tb}^{k,j}\mathbf{x}_b^j + \mathbf{n}_{tb} \\ &= \mathbf{X}_{ntb}\boldsymbol{\alpha}_{ntb}^k + \mathbf{n}_{tb}, \quad \text{target present} \end{aligned} \tag{11}$$

where

$$\begin{aligned} \mathbf{X}_{ntb} &= \begin{bmatrix}\mathbf{x}_t & \mathbf{x}_b^1 & \cdots & \mathbf{x}_b^{n_b} & \mathbf{x}_t \odot \mathbf{x}_b^1 & \cdots & \mathbf{x}_t \odot \mathbf{x}_b^{n_b}\end{bmatrix} \\ \boldsymbol{\alpha}_{ntb}^k &= \begin{bmatrix}\alpha_t^k & \alpha_{b,1}^{k,1} & \cdots & \alpha_{b,1}^{k,n_b} & \beta_{tb}^{k,1} & \cdots & \beta_{tb}^{k,n_b}\end{bmatrix}^T. \end{aligned} \tag{12}$$

In general, one mixed pixel only includes part of $\mathbf{X}_{ntb}$, which means that most of entries in $\boldsymbol{\alpha}_{ntb}^k$ are zero. On the basis of this cogitation, we further restrain $\boldsymbol{\alpha}_{ntb}^k$ to be sparse with ℓ_1-norm, and the objective functions for estimating $\boldsymbol{\alpha}_{b,0}^k$ and $\boldsymbol{\alpha}_{ntb}^k$ are given as follows:

$$\hat{\boldsymbol{\alpha}}_{b,0}^k = \arg\min_{\boldsymbol{\alpha}_{b,0}^k}\left\{\frac{1}{2}\left\|\mathbf{x}^k - \mathbf{X}_b\boldsymbol{\alpha}_{b,0}^k\right\|_2^2\right\} \tag{13a}$$

$$\hat{\boldsymbol{\alpha}}_{ntb}^k = \arg\min_{\boldsymbol{\alpha}_{ntb}^k}\left\{\frac{1}{2}\left\|\mathbf{x}^k - \mathbf{X}_{ntb}\boldsymbol{\alpha}_{ntb}^k\right\|_2^2 + \lambda\left\|\boldsymbol{\alpha}_{ntb}^k\right\|_1\right\} \tag{13b}$$

in which λ is the sparsity parameter. It is obvious that $\hat{\boldsymbol{\alpha}}_{b,0}^k$ can be calculated via (6a), and the way to optimize the subproblem (13b) is presented in next subsection.

2) Optimization: It can be observed from formula (13b) that this subproblem is in the form of a mean square error (MSE) function and an ℓ_1-norm regularization term and may be solved via the alternating direction method of multipliers (ADMM) [36]. ADMM first splits the problem into two separated parts and then optimizes them alternatively. For problem (13b), we introduce a vector $\mathbf{u}$ as an auxiliary variable and rewrite the problem in the following form:

$$\begin{aligned} \hat{\boldsymbol{\alpha}}_{ntb}^k = \arg\min_{\boldsymbol{\alpha}_{ntb}^k}&\left\{\frac{1}{2}\left\|\mathbf{x}^k - \mathbf{X}_{ntb}\boldsymbol{\alpha}_{ntb}^k\right\|_2^2 + \lambda\|\mathbf{u}\|_1\right\} \\ &\text{s.t. } \boldsymbol{\alpha}_{ntb}^k = \mathbf{u} \end{aligned} \tag{14}$$

by using the Lagrangian function; (14) is further rewritten as

$$\begin{aligned} L\left(\boldsymbol{\alpha}_{ntb}^k, \mathbf{u}, \mathbf{p}\right) = &\frac{1}{2}\left\|\mathbf{x}^k - \mathbf{X}_{ntb}\boldsymbol{\alpha}_{ntb}^k\right\|_2^2 + \lambda\|\mathbf{u}\|_1 \\ &+ \frac{\mu}{2}\left\|\boldsymbol{\alpha}_{ntb}^k - \mathbf{u} + \frac{1}{\mu}\mathbf{p}\right\|_2^2 \end{aligned} \tag{15}$$

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where $\mathbf{p}$ denotes the Lagrangian matrix, and μ is a positive scalar initialized as 10^{-3} in this article. ADMM alternatively updates $\boldsymbol{\alpha}_{ntb}^{k}$, $\mathbf{u}$, and $\mathbf{p}$. Specifically,

Update $\boldsymbol{\alpha}_{ntb}^{k}$*:* The solution for $\boldsymbol{\alpha}_{ntb}^{k}$ is given as follows:

$$\begin{aligned}\boldsymbol{\alpha}_{ntb}^{k\ (m+1)} &= \arg\min_{\boldsymbol{\alpha}_{ntb}^{k\ (m)}} \left\{ \frac{1}{2}\left\|\mathbf{x}^k - \mathbf{X}_{ntb}\boldsymbol{\alpha}_{ntb}^{k\ (m)}\right\|_2^2 + \frac{\mu^{(m)}}{2} \right. \\ &\qquad \left. \times \left\|\boldsymbol{\alpha}_{ntb}^{k\ (m)} - \mathbf{u}^{(m)} + \frac{1}{\mu^{(m)}}\mathbf{p}^{(m)}\right\|_2^2 \right\} \\ &= \left(\mathbf{X}_{ntb}^T\mathbf{X}_{ntb} + \mu^{(m)}\mathbf{I}_{(1+2n_b)}\right)^{-1} \\ &\qquad \times \left[\mathbf{X}_{ntb}^T\mathbf{x}^k + \mu^{(m)}\left(\mathbf{u}^{(m)} - \frac{1}{\mu^{(m)}}\mathbf{p}^{(m)}\right)\right]. \end{aligned} \tag{16}$$

Update $\mathbf{u}$*:* The solution for $\mathbf{u}$ is derived by

$$\begin{aligned}\mathbf{u}^{(m+1)} &= \arg\min_{\mathbf{u}^{(m)}} \left\{ \lambda\left\|\mathbf{u}^{(m)}\right\|_1 \right. \\ &\qquad \left. + \frac{\mu^{(m)}}{2}\left\|\boldsymbol{\alpha}_{ntb}^{k\ (m+1)} - \mathbf{u}^{(m)} + \frac{1}{\mu^{(m)}}\mathbf{p}^{(m)}\right\|_2^2 \right\} \\ &= S_{\lambda/\mu^{(m)}}\left(\boldsymbol{\alpha}_{ntb}^{k\ (m+1)} + \frac{1}{\mu^{(m)}}\mathbf{p}^{(m)}\right) \end{aligned} \tag{17}$$

where

$$S_\kappa(z) = \begin{cases} z-\kappa, & z > \kappa \\ 0, & |z| \le \kappa \\ z+\kappa, & z < -\kappa \end{cases}$$

is the soft threshold operator and operates elementwise if z is a vector.

Update $\mathbf{p}$*:* The Lagrangian matrix $\mathbf{p}$ is updated as follows:

$$\mathbf{p}^{(m+1)} = \mathbf{p}^{(m)} + \mu^{(m)}\left(\boldsymbol{\alpha}_{ntb}^{k\ (m+1)} - \mathbf{u}^{(m+1)}\right). \tag{18}$$

Finally, μ is updated by a fixed multiplication factor, defined as $\mu^{(m+1)} = 1.05\mu^{(m)}$. The aforementioned optimization process is repeated until $\|\boldsymbol{\alpha}_{ntb}^{k\ (m)} - \mathbf{u}^{(m)}\|_2^2 < \epsilon$ and $\|\mathbf{u}^{(m)} - \mathbf{u}^{(m-1)}\|_2^2 < \epsilon$ (ϵ is set as 10^{-6} here).

B. Class-Guided Automatic Background Generation (CABG)

The ATGP is frequently used for endmember extraction in hyperspectral unmixing. However, it selects pixels with the maximum energy in the orthogonal complement subspace as endmembers and is sensitive to noise [37]. In this article, we propose a CABG method to build a pure background subspace as much as possible. In order to avoid wrong selection of noisy and boundary pixels as background, CABG selects background according to three principles: 1) large distinction with target prior $\mathbf{x}_t$; 2) large density obtained by the density peak clustering (DPC) algorithm [38]; and 3) no class duplication between the extracted background samples. Given an HSI, we first obtain the density value and class information and then carry out CABG to construct background subspace.

1) Density-Based Coarse Classification: DPC is a highly efficient clustering procedure. In this work, noisy and boundary pixels are first detected, and then, the remaining core pixels are labeled according to their density characteristics. The whole process includes local density calculation for each pixel, noisy and boundary pixel detection, and core pixel classification.

For the kth pixel $\mathbf{x}^k \in \mathbb{R}^{n_l\times 1}$, there are two widely used kernels to calculate its local density, defined as follows:

$$\rho^k = \sum_{l=1}^{n_h\times n_w} \chi\left(d^{k,l} - d^c\right) \tag{19a}$$

$$\rho^k = \sum_{l=1}^{n_h\times n_w} e^{-\left(\frac{d^{k,l}}{d^c}\right)^2} \tag{19b}$$

where $\chi(\cdot)$ is the Heaviside step function, $d^{k,l}$ represents the distance between pixels $\mathbf{x}^k$ and $\mathbf{x}^l$, and d^c is a cutoff distance. In this article, a spatial–spectral distance (S^2D) inspired by [39] is applied to better measure similarity/dissimilarity between hyperspectral signatures (see the Appendix for details).

As d^c selection is greatly affected by the distance measurement, a kNN-based kernel based on [40] is applied to address this issue, expressed as follows:

$$\rho^k = \max\left\{1 - \frac{1}{n_k}\sum_{\mathbf{x}^l\in \mathrm{kNN}\left(\mathbf{x}^k\right)} d^{k,l}, 0\right\} \tag{20}$$

where n_k represents the number of nearest neighbors, set as 10% of the number of pixels in HSI.

In general, a core pixel has much more nearest neighbors with small distances than noise/border does and, thus, gets larger density according to (20). This characteristics could be exploited to separate noise/border and reliable background candidates. Specifically, the noise and boundary pixels' detection is formulated as

$$\mathbf{x}^k \text{ is } \begin{cases} \text{noise / border}, & \rho^k \le (\bar{\rho} - \rho^\sigma) \\ \text{background candidates}, & \rho^k > (\bar{\rho} - \rho^\sigma) \end{cases} \tag{21}$$

where $\bar{\rho}$ and ρ^σ are the mean and standard derivation density value of all pixels, respectively. It should be noted that the target of interest usually occupies a small proportion of HSI in HTD and has the risk of being misclassified as noise/border due to its small density. However, it has little negative effects on pure background extraction since our goal is to obtain reliable background candidates, and the misclassification of target into noise/border further enhances background purity.

Moreover, in order to avoid generating background samples belonging to the same category, we further classify the background candidates in an unsupervised manner. As a cluster center should have a large density but also be far from other centers, DPC presented a delta value to measure the distance between pixel $\mathbf{x}^k$ and other pixels, which may be a cluster center with a higher probability than $\mathbf{x}^k$, denoted as $\{\mathbf{x}^l | \rho^l > \rho^k\}$. This article applies the same method, and delta is calculated as follows:

$$\delta^k = \begin{cases} \min\limits_{\mathbf{x}^l:\rho^l>\rho^k} d^{k,l}, & \left\{\mathbf{x}^l | \rho^l > \rho^k\right\} \neq \emptyset \\ \max\limits_{\forall \mathbf{x}^l} d^{k,l}, & \left\{\mathbf{x}^l | \rho^l > \rho^k\right\} = \emptyset. \end{cases} \tag{22}$$

With the local density and delta value, pixels with large density and delta values are selected as cluster centers.

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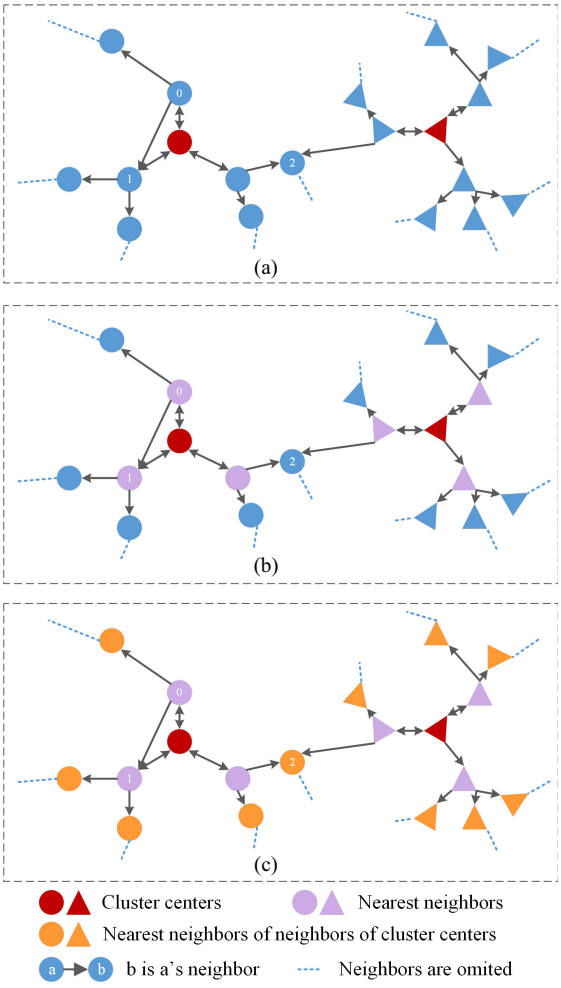

Fig. 1. Illustration of background candidates' classification in a propagated way. (a) Labeling cluster centers. (b) Labeling nearest neighbors of cluster centers. (c) Labeling nearest neighbors of neighbors' cluster centers.

The following score is given to determine the probability of a pixel to be a cluster center:

$$s^k = \left(\frac{\rho^k}{\rho^{\max}}\right)^2 \frac{\delta^k}{\delta^{\max}}, \quad k \in \left\{l \,|\, \rho^l > (\bar{\rho} - \rho^\sigma)\right\}. \tag{23}$$

The pixels with larger s^k are likely to be a cluster center, and we initialize $3n_{vd}$ cluster centers according to the ranking of s^k, where n_{vd} is determined by Harsanyi–Farrand–Chang (HFC) [41] under a probability parameter (set to 0.001 in this article). The background candidate classification is implemented in a propagated way. Take a cluster center $\mathbf{x}_c$ as an example: the whole process begins from labeling its nearest neighbors kNN($\mathbf{x}_c$) and then propagates to the nearest neighbors of samples belonging to kNN($\mathbf{x}_c$). Fig. 1 visually describes this process. There are three special samples marked as "0," "1," and "2," respectively. It can be found that we need to relabel sample "1" since it is a neighbor of sample "0." There are two options: if sample "1" and sample "0" belong to the same class, i.e., exactly the case displayed in Fig. 1, skip this step; otherwise, label pixel "1" the same as the one (pixel "0" or the cluster center of the class of pixel "1") that is more similar to pixel "1." For pixel "2," adjacent to two pixels with different classes, we label it by comparing its similarity with these two pixels.

2) CABG: With the coarse classification map, we extract representative background samples from the background candidates with high purity and diversity. The pixels that have the maximum distinction with target prior $\mathbf{x}_t$ and large density values are selected as background. Specifically, we first compute the orthogonal subspace projection distance between $\mathbf{x}_t$ and other background candidates as follows:

$$d_{\text{sp}}^{k(1)} = \left(\mathbf{I}_{n_l} - \mathbf{x}_t\left(\mathbf{x}_t^T\mathbf{x}_t\right)^{-1}\mathbf{x}_t^T\right)\mathbf{x}^k, \quad k \in \left\{l \,|\, \rho^l > (\bar{\rho} - \rho^\sigma)\right\} \tag{24}$$

then sort $\mathbf{d}_{\text{sp}}^{(1)} = [d_{\text{sp}}^{1(1)} \; d_{\text{sp}}^{2(1)} \; \ldots \; d_{\text{sp}}^{|l|(1)}]$ in a descent order, the corresponding index vector is denoted as $\mathbf{d}_{\text{idx}}$, and $|l|$ is the number of elements in set $\{l|\rho^l > (\bar{\rho} - \rho^\sigma)\}$, i.e., the number of background candidates. The first background sample is obtained according to the following criteria:

$$\mathbf{x}_b^{(1)} = \arg\max_{\mathbf{x}^k} \left\{\rho^k \,|\, k \in \mathbf{d}_{\text{idx}}[1:n_d], c\left(\mathbf{x}^k\right) \neq c(\mathbf{x}_t)\right\} \tag{25}$$

where $c(\cdot)$ outputs the class of each sample and n_d is a parameter set to 1% of the total number of pixels in the dataset.

Similarly, the other background samples are obtained by iteratively applying (24) and (25) until there is no other k satisfying the conditions in (25). Note that $\mathbf{x}_t$ should be updated by $[\mathbf{x}_t \; \mathbf{x}_b^{(1)} \; \ldots \; \mathbf{x}_b^{(m-1)}]$ when finding the mth background pixel. Finally, we construct orthogonal background subspace via PCA, and the number of leading eigenvectors is determined by n_{vd}.

It should be noted that no class information of HSI is required in advance, and the coarse classification by DPC performs in an unsupervised way. The target prior and background extracted by CABG are used as training samples, and the remaining pixels in HSI are used as test samples, whose detection values can be calculated according to Section III-A.

IV. Experimental Results

A. Experimental Settings

1) Datasets: The effectiveness of the proposed BSTD is evaluated on five real-world HSIs, the detailed information of which is provided in the following.

1) *San Diego:* The first and second datasets are acquired by Airborne Visible/InfraRed Imaging Spectrometer (AVIRIS) ranging from 0.4 to 2.5 μm with an interval of 10 nm and cover over San Diego airport, California, USA, with a spatial resolution of 3.5 m. The two study subimages, named San Diego I and San Diego II, occupy 100 × 100 pixels and remain 189 informative bands after removing water absorption and low signal-to-noise ratio (SNR) bands. Each subimage includes three parked airplanes as targets of interest, whose location is marked in Fig. 2.
2) *Avon:* The third dataset is collected by a hyperspectral pushbroom ProSpecTIR-VS sensor, including 360 spectral bands covering wavelengths from 0.4 to 2.45 μm [42], and its spectral resolution is about 5 nm.

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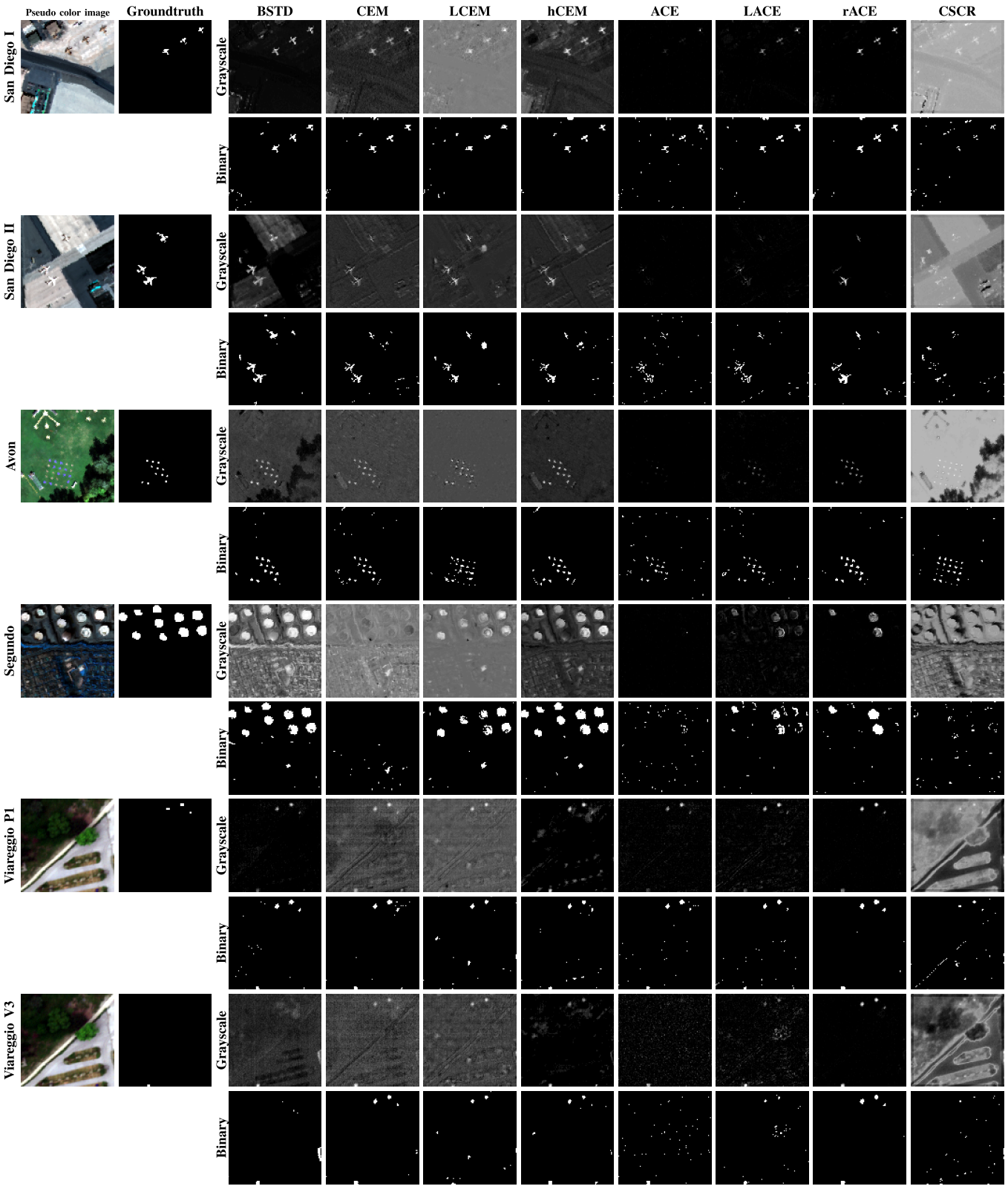

Fig. 2. Pseudocolor image of HSIs, manually annotated ground truth, gray-scale detection maps, and binary detection maps (P_f is fixed to 0.005) obtained by different detection methods.

This image was taken over Avon, New York, USA, on September 20, 2012, and its spatial size is 100×100 with 1 m/pixel resolution. There are two colors of blue and brown tarps arranged in the scene, and the blue tarp is selected as the target of interest to be detected. Fig. 2 displays the pseudocolor image and ground truth.

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3) *Segundo:* The fourth dataset is also acquired by AVIRIS over El Segundo, California, USA. The spectral range is the same as San Diego, but 211 bands are preserved after useless bands removal. Different from San Diego, the spatial resolution of Segundo is 7.1 m probably due to the flight height. This image is composed of 100×100 pixels with several storage tanks chosen as the main observation target, as shown in Fig. 2.
4) *Viareggio [43]:* The fifth dataset is captured by a push-broom hyperspectral Sistema Iperspettrale Modulare. The Galileo Avionica (SIM.GA) sensor over Viareggio, Tuscany, Italy, on May 8, 2013. The spatial resolution and the spectral resolution are 0.6 m and 2.3 nm covering wavelengths from 0.4 to 1.0 μm, respectively. The size of the applied image is $100 \times 100 \times 127$ by averaging the data over groups of four consecutive spectral channels. There are two kinds of targets deployed in this scene, including three green panels (P1) and one green vehicle (V3), the locations of which are shown in Fig. 2.

2) Evaluation Metrics: In addition to visualized gray-scale and binary detection maps, objective evaluation indicators should also be applied for fair performance comparison.

1) *Receiver Operating Characteristic (ROC) Curve:* The ROC analysis has been widely used for effectiveness verification in the binary classification task. It is specified by false alarm rate P_f and detection probability P_d, corresponding to the horizontal axis and the vertical axis, respectively. It is obvious that a good detection approach should simultaneously have high P_d and low P_f, i.e., the better the detector, the closer the ROC curve is to the upper left of the coordinate axis.
2) *Area Under ROC ($AUC_{(P_f,P_d)}$):* Considering that ROC curves of different methods might intersect with each other, it is difficult to determine which detector is superior. To overcome this issue, $AUC_{(P_f,P_d)}$ is developed to calculate the area under each ROC curve. The method with higher $AUC_{(P_f,P_d)}$ is assessed to yield more promising detection performance.
3) *Minimal P_f at Full Detection ($P_f^{min}(P_d = 1)$):* In some applications, such as medical diagnosis and security inspection, people are more concerned about whether all positive samples are correctly detected at an acceptable false alarm rate. Therefore, $P_f^{min}(P_d = 1)$ is presented in response to this situation. An outstanding detector should achieve full detection at a lower false alarm rate.

3) Parameter Selection: In BSTD, there are two important parameters: the spatial size of 3-D patch n_p for spatial–spectral distance calculation (see the Appendix) and the sparsity parameter λ in formula (13b). This subsection gives a detailed analysis of their impacts on detection performance and provides a few guidelines about parameter setup.

1) n_p*:* In general, large n_p could alleviate the negative influences of spectral variation on distance measurement. However, too large n_p would introduce interferential neighboring information and lead to inaccurate similarity measurement and a dramatic increase in the computational costs. Conversely, small n_p cannot provide sufficient spatial information. In order to obtain appropriate n_p, we determine its value referring to the target size. In the experiments, n_p is set to 7, 7, 5, 9, 5, and 5 for San Diego I, San Diego II, Avon, Segundo, Viareggio P1, and Viareggio V3, respectively.
2) λ*:* To analyze the effects of λ on detection performance, experiments on five hyperspectral datasets are carried out under different values of λ, i.e., 0.0001, 0.0005, 0.001, 0.005, 0.01, 0.05, 0.1, 0.5, and 1.0. Fig. 3 shows $AUC_{(P_f,P_d)}$ values of BSTD for the test HSIs as λ varies from 0.0001 to 1.0. For clarity, we separately normalize the $AUC_{(P_f,P_d)}$ values. It can be observed that BSTD generally exhibits satisfactory performance at low λ, and the detection accuracy gradually decreases as λ increases. On the basis of the experimental results, λ is recommended to be set to 0.001.

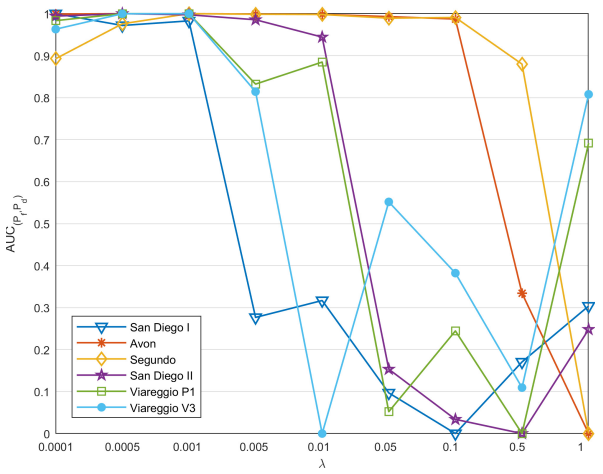

Fig. 3. Impacts of λ on detection accuracy $AUC_{(P_f,P_d)}$ for all test HSIs.

B. Comparison

1) Component Analysis: The component analysis is performed to explain the effects of the nonlinear mixture and sparse abundance on target detection. The linear mixture-based AMSD is applied as the backbone, and MSDinter is extended from AMSD by introducing a nonlinear mixture, while BSTD further takes sparse characteristics into account. For a convenient comparison, the linear background subspace used in AMSD and MSDinter is the same as the one in BSTD, instead of the dual-window strategy as in [26].

Fig. 4 presents the detection accuracy values ($AUC_{(P_f,P_d)}$ and $P_f^{min}(P_d = 1)$) of AMSD, MSDinter, and BSTD on five datasets. In comparison with AMSD, MSDinter performs far better on Avon, Viareggio P1, and Viareggio V3 but slightly worse on San Diego I, San Diego II, and Segundo, which may depend on the type of spectral mixture. It can be ascertained that there are some taller trees in Avon and Viareggio, and the light reaching the sensor may be easily scattered by these trees and then reflected by artificially placed targets. This is precisely the case of bilinear interactions and regularly occurs in forested areas [44]. Therefore, MSDinter could achieve desired results in this situation but would be seriously hindered

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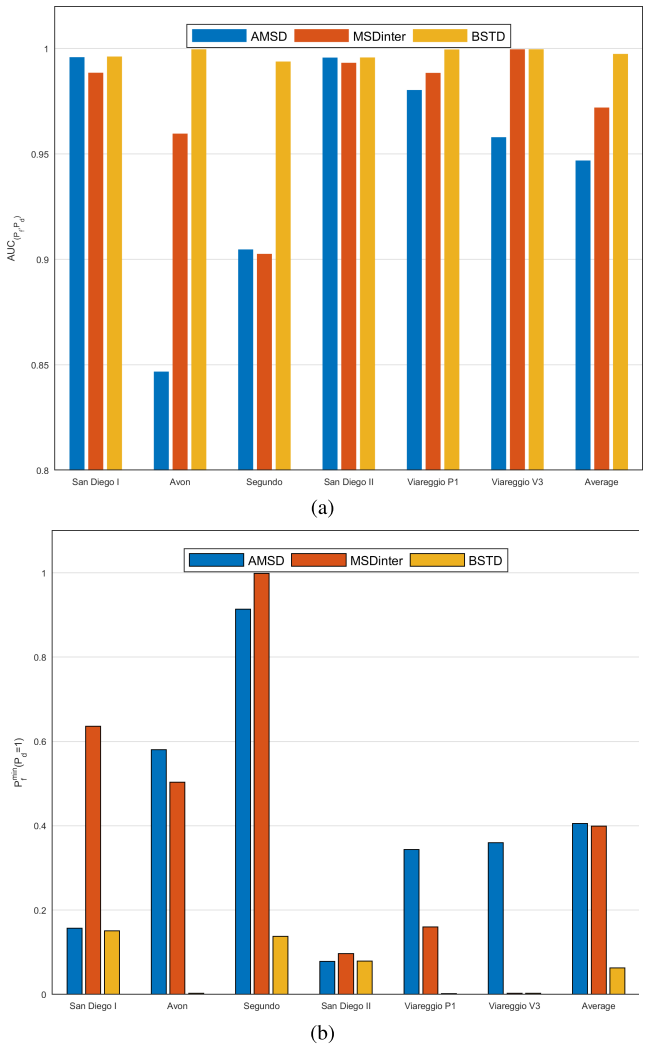

Fig. 4. Detection accuracies, including $AUC_{(P_f,P_d)}$ and $P_f^{min}(P_d = 1)$ of AMSD, MSDinter, and BSTD for all test cases, as well as their averaged accuracy. (a) $AUC_{(P_f,P_d)}$. (b) $P_f^{min}(P_d = 1)$.

if the data mixture of HSI is inconsistent with BMM. Based on this characterization, MSDinter could be used as a tool for judging target–background mixture types to a certain extent. Although both MSDinter and BSTD are established on BMM, BSTD attains an improvement of detection accuracy on San Diego I, San Diego II, and Segundo datasets since it can automatically extract beneficial linear and bilinear features by employing the sparsity constraint and, thus, avoid performance degradation caused by inappropriate bilinear signatures. Moreover, it should be noticed that the improvement of BSTD to AMSD on San Diego I and San Diego II is not very apparent, implicitly indicating that the mixture between target and background is likely to be linear. As for Segundo, it can be preliminarily stated that the mixture model may be an unknown nonlinear mixture model rather than BMM because the slight superiority of BSTD to AMSD implies not only linear mixture, and the inferiority of MSDinter to AMSD implies a more complicated mixture model than BMM.

Moreover, we locate the background pixels extracted by ATGP and CABG on the pseudocolor images for San Diego I, San Diego II, Avon, Segundo, Viareggio P1, and Viareggio V3, as shown in Fig. 5. In order to evaluate background diversity, the SAM (radian) values between every two background samples are calculated, and the minimum SAM (denoted as SAM_{min}) is used for diversity evaluation. In comparison with ATGP, CABG could achieve higher SAM values in most cases, indicating better background diversity. On the other hand, we notice that CABG extracts fewer background pixels than ATGP since it filters out some highly similar samples based on the guided classification map and, thus, helps to generate background samples with higher diversity.

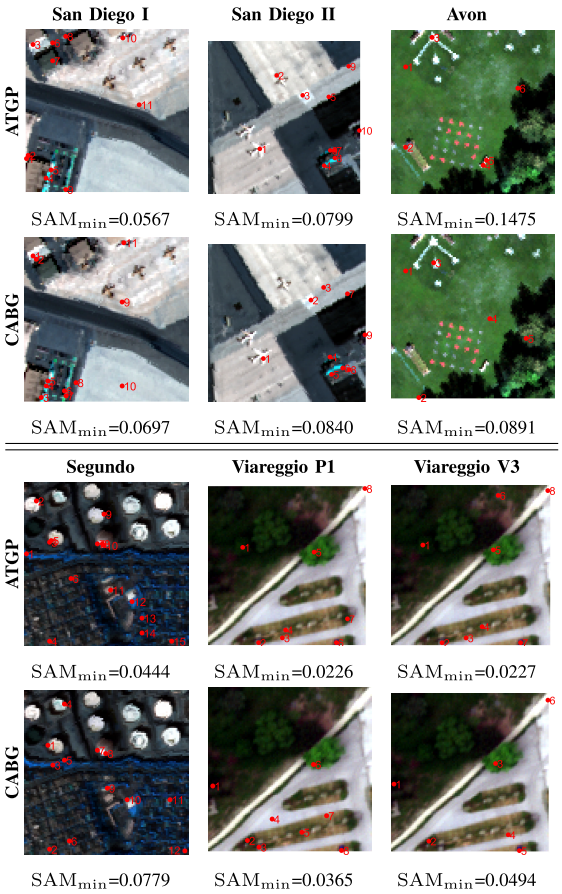

Fig. 5. Location of background samples extracted by ATGP and CABG.

2) Detection Statistics: This subsection provides performance evaluation of BSTD by comparing with constrained energy minimization (CEM) [45], local CEM (LCEM), hierarchical CEM (hCEM) [46], ACE, local ACE (LACE), reweighted ACE (rACE) [47], and CSCR. The prior knowledge demanded by CEM and ACE is only the spectral information of the target to be detected, which is kept consistent for all methods. For San Diego I, San Diego II, Avon, and Segundo, the target spectrum that is most similar to the average spectrum of all target pixels in the image is selected as the required target prior. For Viareggio P1 and Viareggio V3, the spectral information of the target is acquired by the hand-held FieldSpec spectroradiometer. For hCEM, there are two important parameters λ and ϵ set to 200 and 0.05 for all datasets to avoid excessive background suppression. For

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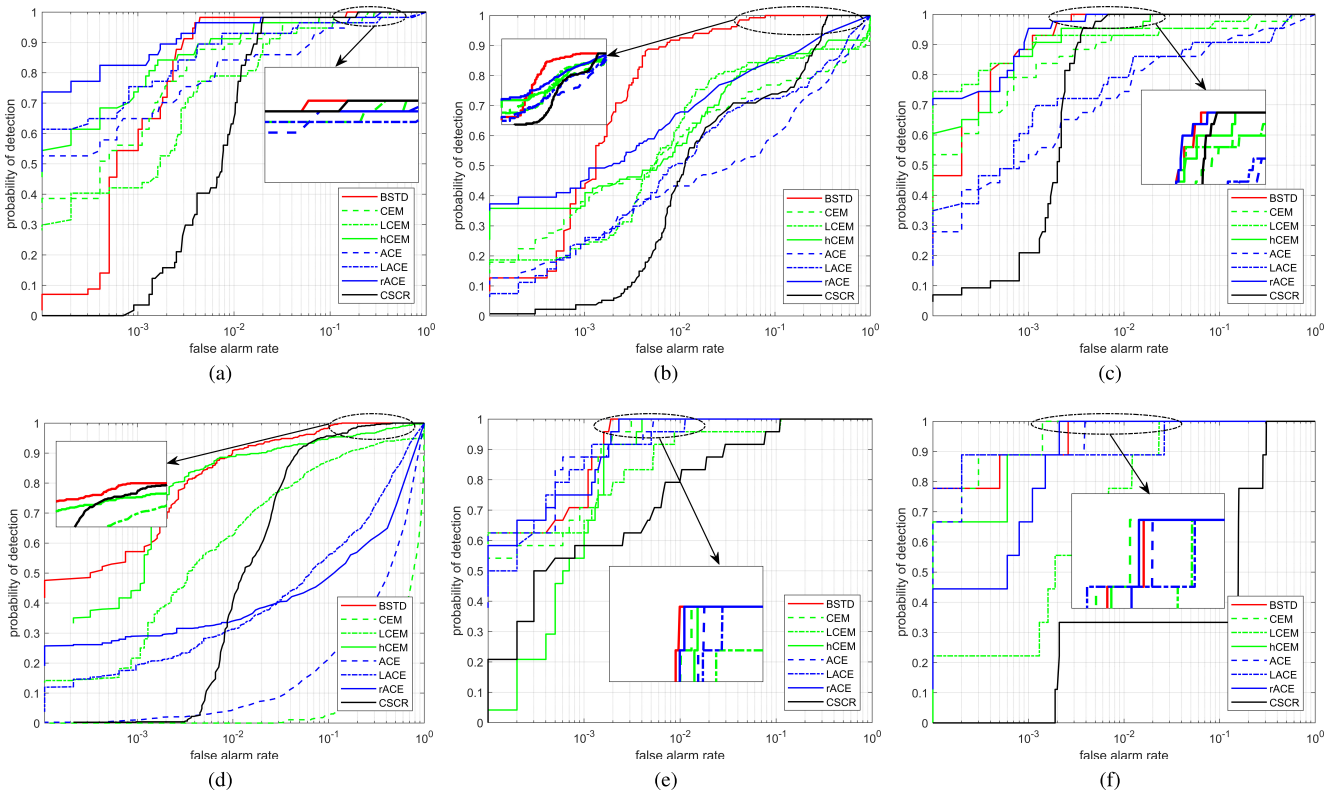

Fig. 6. ROC curves for different compared methods on six HSIs. For easy comparison, the horizontal axis is in a logarithmic scale, and the area that P_d reaches 100% for the first time is enlarged. (a) San Diego I. (b) San Diego II. (c) Avon. (d) Segundo. (e) Viareggio P1. (f) Viareggio V3.

rACE, we adjust ϵ (different from ϵ in hCEM) to perform detection for different images, and rACE reaches the best accuracy when ϵ is equal to 10^{-3}, 10^{-3}, 10^{-2}, 10^{-3}, 10^{-3}, and 10^{-3} for San Diego I, San Diego II, Avon, Segundo, Viareggio P1, and Viareggio V3, respectively. For CSCR, the inner and outer window sizes, denoted as (w_i, w_o), are determined according to the size of target, setting to $(5, 11)$, $(5, 11)$, $(3, 7)$, $(7, 13)$, $(3, 9)$, and $(3, 9)$ on the experimental datasets. For each test pixel in LCEM/LACE, the background for correlation/covariance matrix construction is obtained by the widely used sliding dual-window, i.e., the pixels falling inside the outer window and outside the inner window. For San Diego I, Avon, Viareggio P1, and Viareggio V3, the window sizes of LCEM and LACE are set the same as the one of CSCR. For San Diego II, the dual-window sizes of LCEM and LACE are set as $(15, 25)$ and $(13, 21)$, respectively. For Segundo, the inner and outer window sizes of LCEM and LACE are the same, selected as $(15, 25)$.

1) *Subjective Evaluation:* Fig. 2 visualizes the gray-scale and binary detection maps with $P_f = 0.005$ of all competitors on five hyperspectral datasets. LCEM and LACE obtain detection results due to the local background correlation and covariance matrix, but ACE and LACE are more sensitive to spectral variation of target priors than CEM and LCEM, which can be observed from the gray-scale detection maps. hCEM and rACE highlight targets better than the original CEM and ACE since hCEM and rACE resolve the target contamination issue through hierarchical background suppression and iterative target purification, respectively. Apart from basic CEM and ACE, rACE is the best detector for suppressing background but omits some target pixels, especially on Segundo. CSCR performs very poorly on visual target–background distinction in contrast with other detectors. Visually, both BSTD and hCEM correctly pick up most target pixels with remarkable background suppression.
2) *Objective Evaluation I:* Fig. 6(a)–(f) depicts ROC curves obtained by different detection methods on San Diego I, San Diego II, Avon, Segundo, Viareggio P1, and Viareggio P3, respectively. BSTD distinguishes all target pixels at the lowest false alarm rate on all test images except Viareggio V3, as shown in the enlarged areas of ROC curves. Also, the ROC curve of BSTD is above the other ones in a wide range of P_f. rACE generally detects the most pixels when P_f decreases to a certain level, e.g., 10^{-3}, verifying its superior ability on background suppression, but, for Segundo, it cannot realize full detection until $P_f \approx 1$. This ROC analysis is consistent with the visualized detection map. However, due to the intersection of ROC curves, quantitative metrics should be applied to get a deterministic evaluation.
3) *Objective Evaluation II:* Table III tabulates $\mathrm{AUC}_{(P_f,P_d)}$, $P_f^{\min}(P_d = 1)$ and time consumption of all compared methods. For San Diego I, Avon, Viareggion P1, and Viareggio V3, $\mathrm{AUC}_{(P_f,P_d)}$ obtained by some detection methods are comparable, but there is a big difference between their $P_f^{\min}(P_d = 1)$ values. Take San Diego I

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TABLE III
DETECTION ACCURACY AND TIME CONSUMPTION COMPARISON FOR DIFFERENT DETECTORS APPLIED TO THE SIX TEST DATASETS

Index	Detectors	San Diego I	San Diego II	Avon	Segundo	Viareggio P1	Viareggio V3	Average
$\mathrm{AUC}_{(\mathrm{P_f},\mathrm{P_d})}\uparrow$	BSTD	0.99621	0.99577	0.99963	0.99384	0.99956	0.99966	0.99744
	CEM	0.98633	0.83676	0.98022	0.29411	0.99938	0.99979	0.84943
	LCEM	0.98735	0.88146	0.99261	0.92115	0.99425	0.99419	0.96184
	hCEM	0.99237	0.89621	0.99891	0.97589	0.99897	0.99957	0.97699
	ACE	0.97368	0.81450	0.95116	0.56052	0.99938	0.99948	0.88312
	LACE	0.97693	0.86653	0.95618	0.79705	0.99910	0.99705	0.93214
	rACE	0.99370	0.93351	0.99970	0.72182	0.99952	0.99929	0.94126
	CSCR	0.98916	0.92705	0.99802	0.97325	0.98867	0.86392	0.95668
$\mathrm{P_f^{min}}\,(\mathrm{P_d}=1)\downarrow$	BSTD	0.15046	0.07936	0.00281	0.13781	0.00190	0.00260	0.06249
	CEM	0.42221	0.99980	0.57256	1.00000	0.00311	0.00140	0.49985
	LCEM	0.25163	1.00000	0.21834	0.99989	0.11237	0.02332	0.43426
	hCEM	0.28875	0.99970	0.01888	1.00000	0.00401	0.00210	0.38557
	ACE	0.99990	0.99990	0.99980	0.99989	0.00531	0.00390	0.66812
	LACE	0.99990	0.99990	0.57578	0.99989	0.01133	0.02642	0.60220
	rACE	0.34225	0.99990	0.00402	0.99989	0.00231	0.00210	0.39174
	CSCR	0.19572	0.35587	0.00683	0.46633	0.11047	0.30828	0.24058
Time (in seconds)	BSTD	37.2121	44.7094	43.6797	23.2092	31.5354	31.7023	35.3413
	CEM	0.0143	0.0114	0.0230	0.0106	0.0127	0.0084	0.0134
	LCEM	13.3827	20.7947	27.7144	24.7530	11.4731	10.0692	18.0312
	hCEM	0.0731	0.0602	0.1408	0.1916	1.5333	1.9918	0.6651
	ACE	0.2745	0.2047	0.5196	0.2505	0.2305	0.1493	0.2715
	LACE	25.6437	18.4172	56.4228	18.6655	12.1254	16.8452	24.6886
	rACE	0.8613	1.2057	1.6289	1.4768	0.4200	1.0779	1.1118
	CSCR	11.3690	10.0879	8.5885	14.1908	11.0321	9.7912	10.8432

as an example, BSTD and hCEM differ only 0.384% on $\mathrm{AUC}_{(\mathrm{P_f},\mathrm{P_d})}$ but 13.829% on $\mathrm{P_f^{min}}(\mathrm{P_d}=1)$, indicating the superiority of BSTD to hCEM. As for San Diego II and Segundo, hCEM and rACE present a significant accuracy decrease compared with results on other datasets, but BSTD can still achieve stable detection performance. Moreover, call the detection accuracy shown in Fig. 4, AMSD exhibits higher accuracy values than CEM and ACE in most cases due to its better exploitation of background information, indicating the effectiveness of the proposed robust background generation process. Overall, with the support of subjective and objective evaluation results, BSTD could provide more reliable detection performance than rACE and hCEM. In terms of computation cost, all experiments are carried out in MATLAB R2017b on a laptop with a 1.60-GHz Intel Core i5-8265U CPU and an 8-GB RAM. As shown in Table III, the proposed BSTD is less efficient than other traditional detection methods, mainly due to the class-guided background subspace construction and alternating optimization of ℓ_1-inducing regularization term. We will try to accelerate this method by making improvements in the optimization method and computer configuration in the future.

V. CONCLUSION

This article presents a BHT-based BSTD for hyperspectral datasets, designed by considering bilinear interactions between target and background, and sparse abundance. The sparsity constraint helps extract the most informative linear and bilinear signatures to be used in the subsequent detection procedure. Furthermore, to obtain background subspace with high purity and diversity, a robust automatic background generation method has been designed by excluding noisy and boundary mixed pixels under the guidance of a spatial–spectral density-based classification map. Experimental results on five real-world hyperspectral datasets indicate that the proposed method offers better detection performances than other compared methods and demonstrate the advantages of BSMM and noise-/border-excluded background generation on HTD.

APPENDIX

For n_l-dimensional hyperspectral pixels $\mathbf{x}^k$ and $\mathbf{x}^l$, $\mathrm{S^2D}$ is computed in three steps.

1) Extract two local 3-D patches with the size of $n_p \times n_p \times n_l$ around $\mathbf{x}^k$ and $\mathbf{x}^l$, respectively, denoted as $\mathbf{X}_k$ and $\mathbf{X}^l$, where n_p is the spatial size of 3-D patch.

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2) Compute the distance between 1D-spectra and 3-D patch as follows:

$$\tilde{d}\left(\mathbf{x}^k, \mathbf{X}^l\right) = \begin{cases} \min\limits_{\substack{p=1,\ldots,n_p \\ q=1,\ldots,n_p \\ \mathbf{x}^k \neq \mathbf{x}^l_{p,q}}} d\left(\mathbf{x}^k, \mathbf{x}^l_{p,q}\right), & \mathbf{x}^k \in \mathbf{X}^l \\ \min\limits_{\substack{p=1,\ldots,n_p \\ q=1,\ldots,n_p}} d\left(\mathbf{x}^k, \mathbf{x}^l_{p,q}\right), & \mathbf{x}^k \notin \mathbf{X}^l \end{cases} \tag{26}$$

where $d(\cdot,\cdot)$ could be the Euclidean distance, the spectral angle mapper (SAM), the spectral information divergence (SID), or any other distance metric. SAM is selected in this article. If $\mathbf{x}^k$ is in the spatial window of $\mathbf{X}^l$, we calculate $d(\cdot,\cdot)$ between $\mathbf{x}^k$ and other pixels in $\mathbf{X}^l$ except $\mathbf{x}^k$ to avoid $\tilde{d}(\mathbf{x}^k, \mathbf{X}^l)$ being zero.

3) Obtain S^2D according to the following definition:

$$d^{k,l}_{\mathrm{S^2D}} = d_{\mathrm{S^2D}}\left(\mathbf{x}^k, \mathbf{x}^l\right) = \max\left(\tilde{d}\left(\mathbf{x}^l, \mathbf{X}^k\right), \tilde{d}\left(\mathbf{x}^k, \mathbf{X}^l\right)\right). \tag{27}$$

With the aforementioned definition, it is possible to compute the S^2D value between every pixel pair, denoted by $\mathbf{D}_{\mathrm{S^2D}}$.

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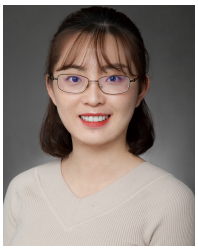
Yanzi Shi (Graduate Student Member, IEEE) received the B.S. degree from the School of Telecommunications Engineering, Xidian University, Xi'an, China, in 2017, where she is pursuing the Ph.D. degree with the State Key Laboratory of Integrated Service Networks.

She is also a Visiting Ph.D. Student with the Telecommunications and Remote Sensing Laboratory, University of Pavia, Pavia, Italy, which is supported by the China Scholarship Council. Her research interests include hyperspectral target detection, feature extraction, deep learning, and urban remote sensing.



Jiaojiao Li (Member, IEEE) received the B.E. degree in computer science and technology, the M.S. degree in software engineering, and the Ph.D. degree in communication and information systems from Xidian University, Xi'an, China, in 2009, 2012, and 2016, respectively.

She was an Exchange Ph.D. Student at Mississippi State University, Starkville, MS, USA. She is a Post-Doctoral Researcher and a Lecturer at Xidian University. Her research interests include hyperspectral remote sensing image analysis and processing, pattern recognition, and data compression.

Yunsong Li (Member, IEEE) received the M.S. degree in telecommunication and information systems and the Ph.D. degree in signal and information processing from Xidian University, Xi'an, China, in 1999 and 2002, respectively.

In 1999, he joined the School of Telecommunications Engineering, Xidian University, where he is a Professor. He is also the Director of the Image Coding and Processing Center, State Key Laboratory of Integrated Service Networks, Xidian University. His research interests focus on image and video processing, hyperspectral image processing, and high-performance computing.

Paolo Gamba (Fellow, IEEE) received the Laurea degree in electronic engineering *(cum laude)* and the Ph.D. degree in electronic engineering from the University of Pavia, Pavia, Italy, in 1989 and 1993, respectively.

He is a Professor with the University of Pavia, where he leads the Telecommunications and Remote Sensing Laboratory. He has been invited to give keynote lectures and tutorials on several occasions about urban remote sensing, data fusion, EO data for physical exposure, and risk management. He has authored or coauthored more than 140 articles in international peer-reviewed journals and presented nearly 300 research works in workshops and conferences.

Dr. Gamba was the Chair of the Data Fusion Committee of the IEEE Geoscience and Remote Sensing Society (GRSS) from October 2005 to May 2009. He has been elected in the GRSS AdCom, since 2014, and he is the GRSS President. He has been the Organizer and the Technical Chair of the Biennial GRSS/ISPRS Joint Workshops on Remote Sensing and Data Fusion over Urban Areas from 2001 to 2015. He also served as the Technical Co-Chair for the 2010 and 2015 IGARSS Conferences, in Honolulu (Hawaii) and Milan (Italy), respectively. He served as an Editor-in-Chief for the IEEE GEOSCIENCE AND REMOTE SENSING LETTERS from 2009 to 2013.

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